Pradhan Mantri Fasal Bima Yojana

(district-level crop insurance dataset)

1. Project Overview and Objective

This project focuses on analyzing the Pradhan Mantri Fasal Bima Yojana (PMFBY) district-level crop insurance dataset using Excel and Power BI. The dataset contains information on insured farmers, area insured, sum insured, premium contributions, farmer categories, and demographic distributions across different states and districts in India.

The main objective of the project is to perform data cleaning and transformation using Excel/Power Query, develop interactive Power BI dashboards, and generate meaningful insights regarding crop insurance coverage, farmer participation, and premium distribution across states, districts, schemes, and seasons.

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2. Data Sources

- **Source Description and Timeline:** India Data Portal 2018-2022
- **Domain**: Agricultural / Crop Insurance

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3. Problem Statement

- To analyze crop insurance participation and coverage under PMFBY across states, districts, seasons, and schemes.
- To identify states and districts with the highest number of insured farmers, insured area, and sum insured.
- To study the contribution of farmers, state governments, and the Government of India towards crop insurance premiums.
- To understand demographic participation based on gender, social categories, and farm-size classifications.

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4. Attribute (Column /Features) Details:

Attribute	Data Type	Description
year	Integer	Year of insurance
season	Text	Kharif/Rabi
scheme	Text	PMFBY/WBCIS
state name	Text	State Name
district name	Text	District Name

Attribute	Data Type	Description
farmer count	Number	Total insured farmers
area insured	Decimal	Total area insured
sum insured	Decimal	Total insured amount
farmer share	Decimal	Farmer premium contribution
Goi share	Decimal	Government of India contribution
state share	Decimal	State Government contribution
male	Decimal	Percentage of male farmers
female	Decimal	Percentage of female farmers

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5. Tools & Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

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6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

- **Data Cleaning & Transformation:**
- Removed duplicate records
- Replaced invalid values (`nan`) with null/appropriate values
- Checked and handled missing values
- Standardized data types
- Filtered and sorted records
- Created Pivot Tables
- Verified district codes and district names
- **Filtering & Sorting:** Organized data to focus on relevant records.
- **Pivot Tables:** Generated Pivot Tables for data summarisation and initial insights.
- Convert the data into Fact and Dimension Table

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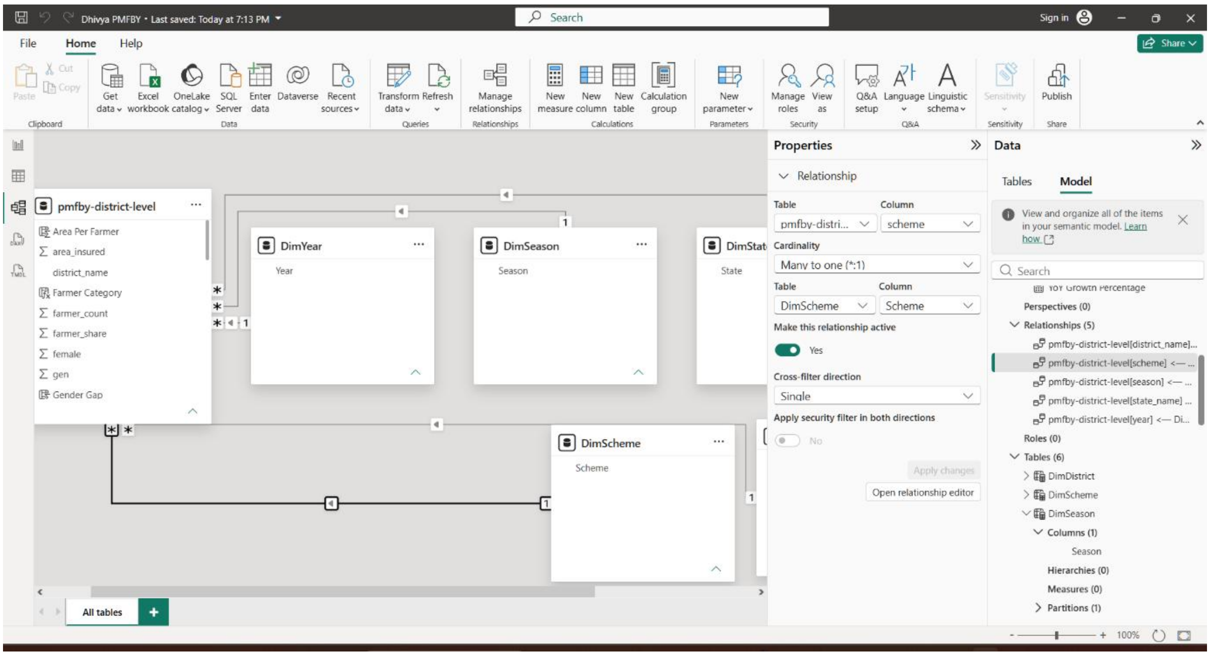
7. Data Modelling and DAX (Power BI)

- **Data Model:** Created a **Star Schema Data Model** with a central Fact table (PMFBY -district-level) and supporting Dimension tables (Dim Year, Dim Season, Dim Scheme, Dim State, Dim District).
- Established **One-to-Many relationships** between dimension tables and the fact table to ensure efficient filtering and analysis.

- Defined appropriate **cardinality and single-direction cross-filtering** to optimize report performance and avoid ambiguity.
- Created **lookup (dimension) tables** using DISTINCT values for Year, Season, Scheme, State, and District to support slicers and improve data organization.



- **Calculated Columns & DAX Measures**
  Implemented DAX measures to calculate key business metrics including Total Farmers Insured, Total Area Covered, Gross Premium, Sum Insured, Government Contribution, State Contribution, Female Participation Percentage, Premium per Farmer, and Year-over-Year Growth Percentage. Calculated columns were created to derive Farmer Categories, Area per Farmer, Gender Gap, and other analytical attributes for enhanced reporting and dashboard insights.

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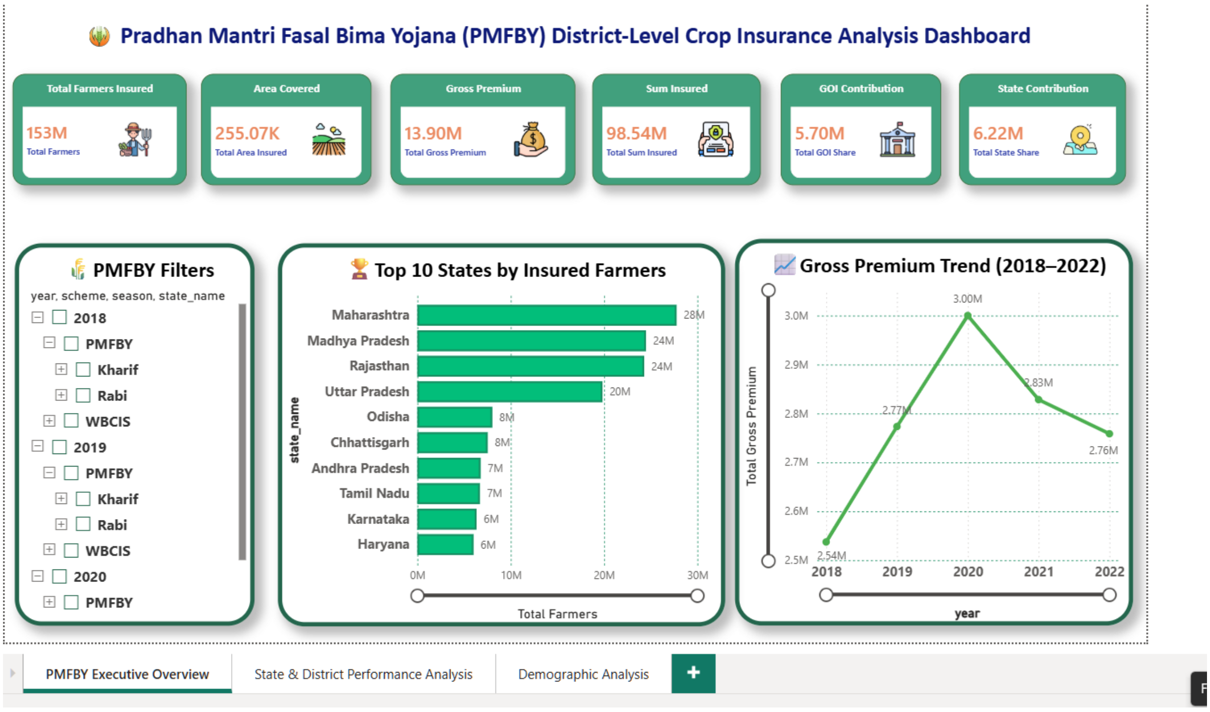
8. Analysis and Visualizations (Power BI)

Dashboard Features

PMFBY Executive Overview

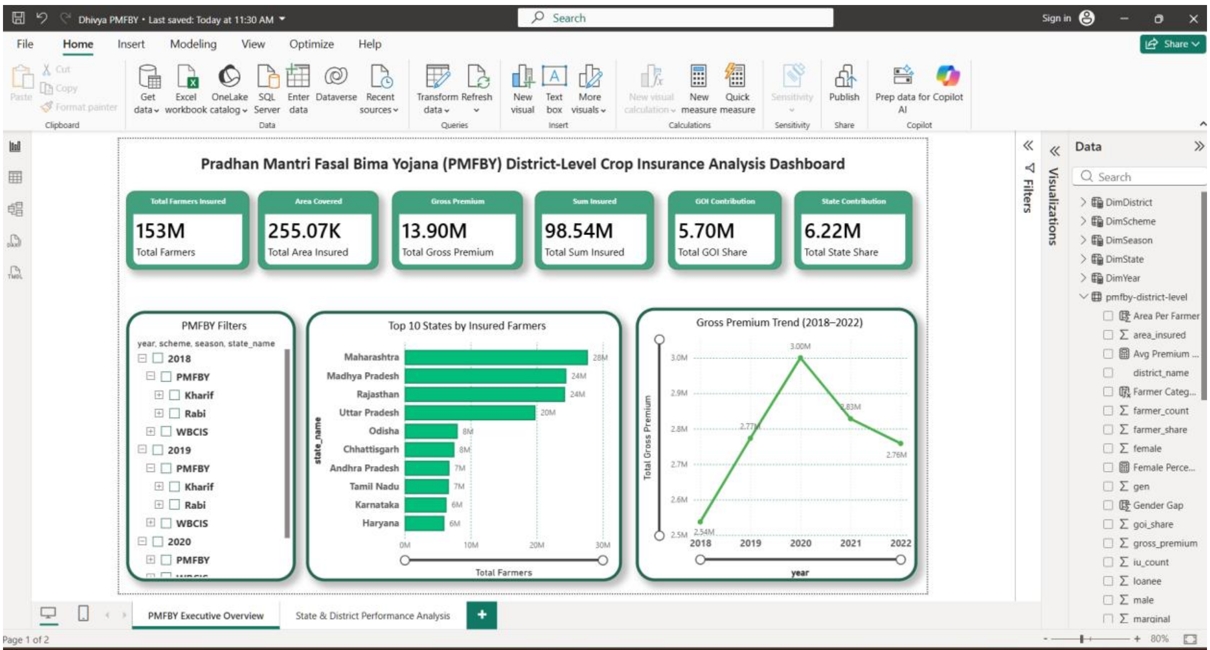


KPI Cards displaying:

- Total Farmers Insured
- Total Area Covered
- Total Gross Premium
- Total Sum Insured
- GOI Contribution
- State Contribution

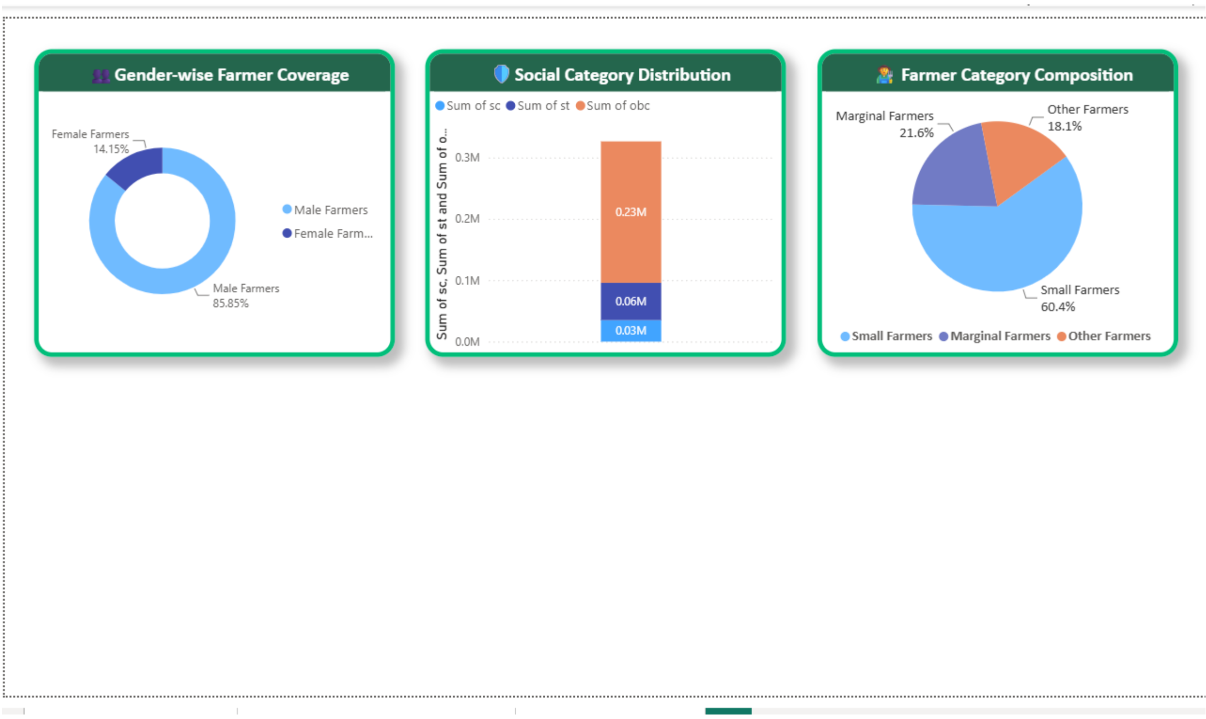
- Bar Chart: Top 10 States by Farmer Coverage
- Line Chart: Gross Premium Trend Analysis (2018–2022)
- Interactive Filter Panel for Year, Scheme, Season, and State selection

State & District Performance Analysis



- Treemap: District-wise Gross Premium Contribution
- Map Visual: Geographical Distribution of Insured Farmers
- Matrix Table: State-wise Coverage & Premium Summary
- KPI Summary Cards: Districts Covered, Farmers Insured, Gross Premium
- Key Insights Panel highlighting major findings

Demographic Analysis



- Donut Chart: Gender-wise Farmer Coverage
- Stacked Column Chart: Social Category Distribution (SC, ST, OBC, Other)
- Pie/Donut Chart: Farmer Category Distribution (Small, Marginal, Other)

Interactive Features Implemented

- Hierarchical Slicers for Year, Scheme, Season, and State
- Cross-filtering and cross-highlighting across visuals
- Drill-down and drill-through capabilities for detailed analysis
- Dynamic filtering using slicers
- Responsive dashboard layout with clear navigation
- Consistent color theme and visual formatting

Bookmarks and Navigation

- Created separate report pages:
  - PMFBY Executive Overview
  - State & District Performance Analysis
  - Demographic Analysis

- Added meaningful visual titles and data labels for better readability
- Implemented page navigation for a seamless user experience

Consolidated Dashboard

A consolidated Power BI dashboard was developed to provide a comprehensive view of PMFBY performance across states, districts, demographic groups, and insurance metrics. The dashboard enables users to monitor farmer participation, premium collection, insured area, government contributions, and demographic coverage through interactive visualizations and filters.

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9. Insights & Conclusions

Descriptive Analysis

- A total of 153 million farmers were insured under PMFBY during the analysis period.
- The total gross premium collected was approximately ₹13.9 million.
- Maharashtra, Rajasthan, and Madhya Pradesh recorded the highest farmer participation.
- Small farmers represented the largest beneficiary group, accounting for over 60% of insured farmers.
- Male farmers constituted the majority of insured beneficiaries.

Diagnostic Analysis

- States with larger agricultural landholdings showed higher PMFBY participation.
- Government premium support encouraged enrollment in major agricultural states.
- Small farmers showed higher dependence on crop insurance due to greater financial risk exposure.
- Premium collections were concentrated in districts with extensive crop cultivation and higher insured area.

Predictive Analysis

- Farmer enrollment is likely to continue growing in states with strong agricultural activity.
- Gross premium collections may increase if insured area expands and more farmers adopt PMFBY.
- States currently showing high participation are expected to remain major contributors.
- Small and marginal farmers will continue to form the largest share of beneficiaries.

Prescriptive Analysis

- Increase awareness programs in low-participation states and districts.

- Encourage greater enrollment of female farmers through targeted outreach initiatives.
- Focus on districts with low insurance penetration to improve coverage.
- Strengthen digital enrollment and claim processing mechanisms.
- Promote PMFBY benefits among small and marginal farmers to increase participation further.

Key Findings

- Maharashtra recorded the highest farmer participation.
- Rajasthan reported the largest insured area.
- Gross premium contributions were concentrated in a limited number of districts.
- Small farmers accounted for the majority of insured beneficiaries.
- Male farmers represented the largest proportion of insured farmers.
- PMFBY coverage remained strongest in major agricultural states.

10. Conclusions

The integration of Excel, Power Query, and Power BI proved effective for conducting end-to-end crop insurance analysis under the Pradhan Mantri Fasal Bima Yojana (PMFBY). Data preprocessing techniques such as cleaning, transformation, and aggregation enabled the creation of a reliable analytical dataset. Power BI data modeling and DAX measures facilitated meaningful performance tracking and interactive reporting.

The dashboard successfully highlighted key trends related to farmer participation, insured area, premium collection, demographic distribution, and state-wise performance. The analysis revealed that farmer coverage and premium contributions were concentrated in a few major agricultural states, while small farmers constituted the largest beneficiary group.

Overall, the project demonstrates how business intelligence tools can transform raw agricultural insurance data into actionable insights, supporting better decision-making, policy evaluation, and program monitoring for crop insurance initiatives.